Interactions in Regression

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Introduction

An interaction term — a product of two predictors entered alongside the main effects — expresses the substantive idea that the effect of one predictor depends on the value of another. Without interactions, a linear model assumes purely additive effects: a unit increase in X_1 produces the same change in Y regardless of X_2. Interactions relax this assumption and often capture the actual scientific hypothesis (treatment effect varies by age, biomarker effect varies by sex, dose-response curve depends on baseline severity). Reporting interactions correctly requires moving beyond main-effect coefficients to conditional effects at specific covariate values.

Prerequisites

A working understanding of multiple linear regression, the geometric meaning of regression coefficients, and centring continuous predictors as a tool for reducing collinearity.

Theory

For two continuous predictors X_1, X_2 and an interaction:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon.$$

The conditional effect of X_1 is $\beta_1 + \beta_3 X_2$ — a function of X_2 rather than a single number. The "main effect" β_1 is the conditional effect of X_1 when $X_2 = 0$; if $X_2 = 0$ is outside the data range or not a meaningful reference, the main-effect coefficient is uninterpretable.

Centring continuous predictors before adding interactions makes $X_2 = 0$ correspond to the mean of X_2 and gives the main-effect coefficient the interpretation "effect at the average of the other predictor". It also reduces the mechanical collinearity between X_1, X_2, and $X_1 X_2$.

For categorical-by-continuous interactions, the continuous coefficient is the effect within the reference category and the interaction is the difference for the non-reference category.

Assumptions

Same as the underlying regression: linearity in the linear predictor (after the interaction is included), independent observations, etc. The interaction itself is a model choice, not an assumption.

R Implementation

```
library(emmeans); library(ggeffects)

d <- mtcars
d$wt_c <- scale(d$wt, center = TRUE, scale = FALSE)[, 1]
d$hp_c <- scale(d$hp, center = TRUE, scale = FALSE)[, 1]

fit <- lm(mpg ~ wt_c * hp_c, data = d)
```

```r
summary(fit)

# Marginal effects at selected levels
emtrends(fit, ~ hp_c, var = "wt_c",
         at = list(hp_c = c(-50, 0, 50)))

# Interaction plot
ggpredict(fit, terms = c("wt_c", "hp_c [-50, 0, 50]")) |> plot()
```

Output & Results

The interaction coefficient is on the product scale. `emtrends()` returns the conditional slope of one variable at chosen values of the other. `ggpredict()` plots predicted Y against X_1 at multiple levels of X_2, the canonical visualisation of an interaction.

Interpretation

A reporting sentence: "The effect of weight on mpg was more negative at high horsepower than at low (interaction coefficient 0.030, 95 % CI 0.007 to 0.053, $p = 0.01$); at the mean horsepower, each 1000-lb increase in weight reduced mpg by 3.5 (95 % CI -4.6 to -2.4); at horsepower 50 above the mean, the slope was -4.9, and at 50 below it was -2.0." Always report simple slopes at meaningful covariate values, not just the interaction coefficient.

Practical Tips

- Centre continuous predictors before adding interactions; this gives interpretable main effects ("effect at the mean of the other") and reduces collinearity between main and interaction terms.
- Never interpret "main effects" when a significant interaction is present without specifying the reference value of the moderator; the main effect alone is incomplete.
- Use `emmeans::emtrends()` or `marginaleffects::slopes()` to compute simple slopes at meaningful covariate values; pair with confidence intervals.
- Visualise interactions with `ggeffects::ggpredict()` plots showing Y against the focal predictor at several values of the moderator; this communicates the interaction far better than coefficient tables.
- For categorical-by-categorical interactions, an interaction plot with separate lines per group is the standard display; for categorical-by-continuous, use facets or colour groups.
- Higher-order interactions (three-way and beyond) require much larger samples and careful interpretation; usually best avoided in confirmatory analyses.

R Packages Used

`emmeans` for marginal means, simple slopes, and contrast-stable post-hoc inference; `ggeffects` for ggplot-style predicted-effect plots; `marginaleffects` for unified marginal-effect computation across many model classes; `interactions` for purpose-built interaction probing tools.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI